Lending Club Case Study

Insights and Recommendations to reduce charge-offs

Goal and Objective

Goal:

The goal is identify attributes of a loan which influence its repayment status i.e to identify which loans are likely to fully paid and which are in the risk of default.

Objective:

- Reduce loss of business to the company by identifying applicants that are likely to repay the loan
- Reduce financial loss by identifying applicants who are not likely to repay the loan

Methodology:

We will use exploratory data analysis to identify borrower and loan attributes that influence the loan repayment status

Understanding the data

1. The loan dataset contains information for 39717 loans aganist 110 features

2. A significant number of columns have missing values. Some of the columns like acc_now_delinq, tot_coll_amt, tot_cur_bal, open_acc_6m, open_il_6m, open_il_12m etc are entirely missing. It indicates that these columns were not filled by the applicant/bank or were not applicable to loans in the data

3. The dataset contains numerical and categorical data. Some numerical data like 'int_rate' might be stored as string objects due to presence of '%'. These might require cleaning to convert them to numerical values for qunatative analysis.

Understanding the data

1. The loan dataset contains information for 39717 loans against 110 features

2. A significant number of columns have missing values. We dropped all columns with more than 75% missing value. We will be working with 54 remaining columns

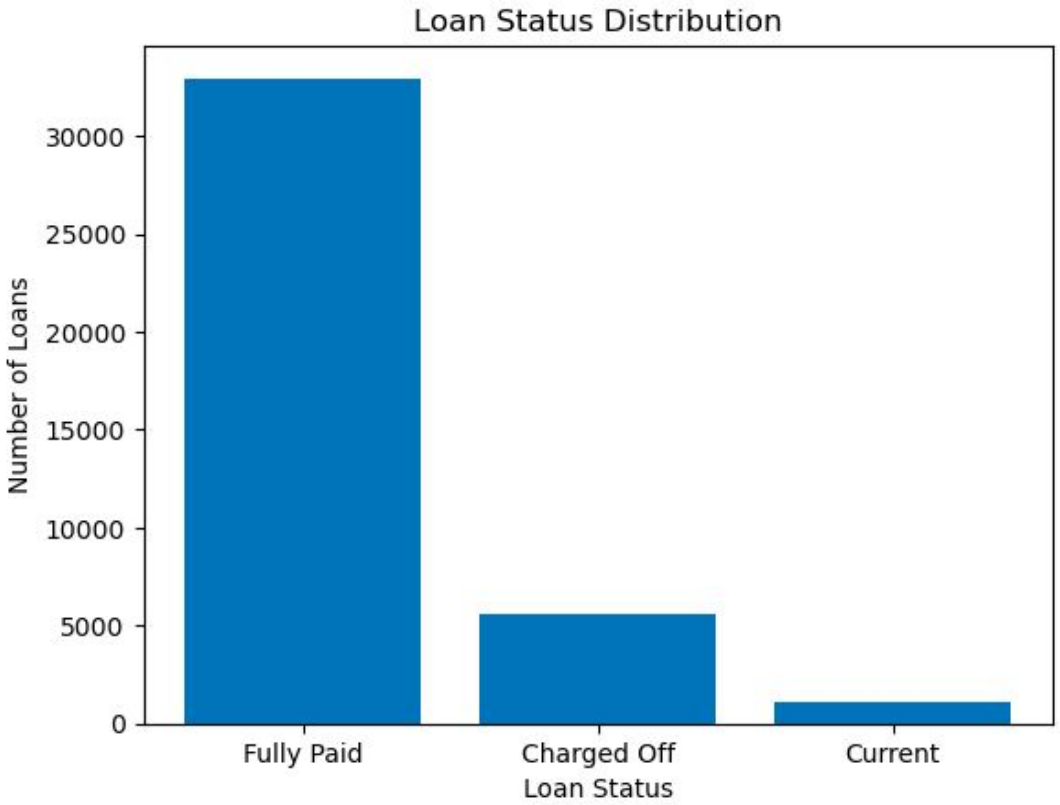
3. The dataset contains numerical and categorical data. Some numerical data like 'int_rate' might be stored as string objects due to presence of '%'. These might require cleaning to convert them to numerical values for quantitative analysis. Similarly, 'issue_d', 'earliest_cr_line', 'last_pymnt_d ', 'last_credit_pull_d' have been converted to the datetime format

3. 'int_rate' which is could be one of the most important driver of loan_status (based on business understanding) has been converted to float64 for quantitative analysis.

4. Some columns like 'revol_util' which are important from business point of view have been Imputed with relevant values values of columns like

Univariate Analysis

Loan Status Distribution



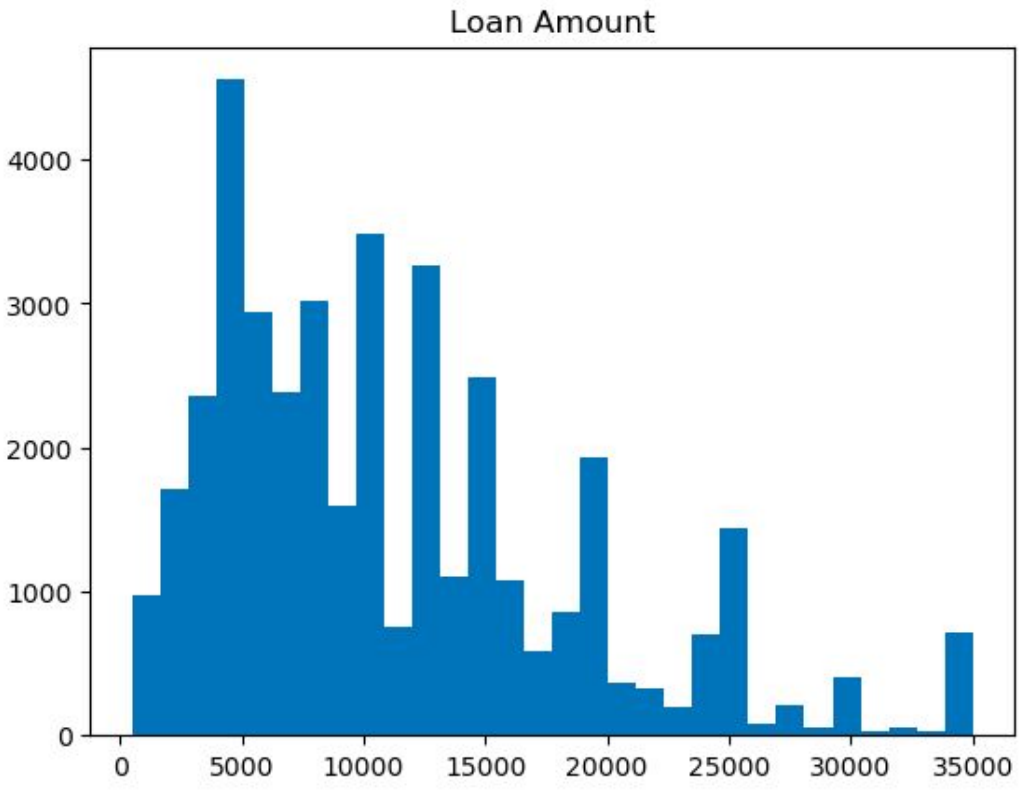
1. Almost 82% of the loan records are paid

2. 14% borrowers have defaulted on their loan repayment

3. ~3% borrowers are currently servicing their loan repayments

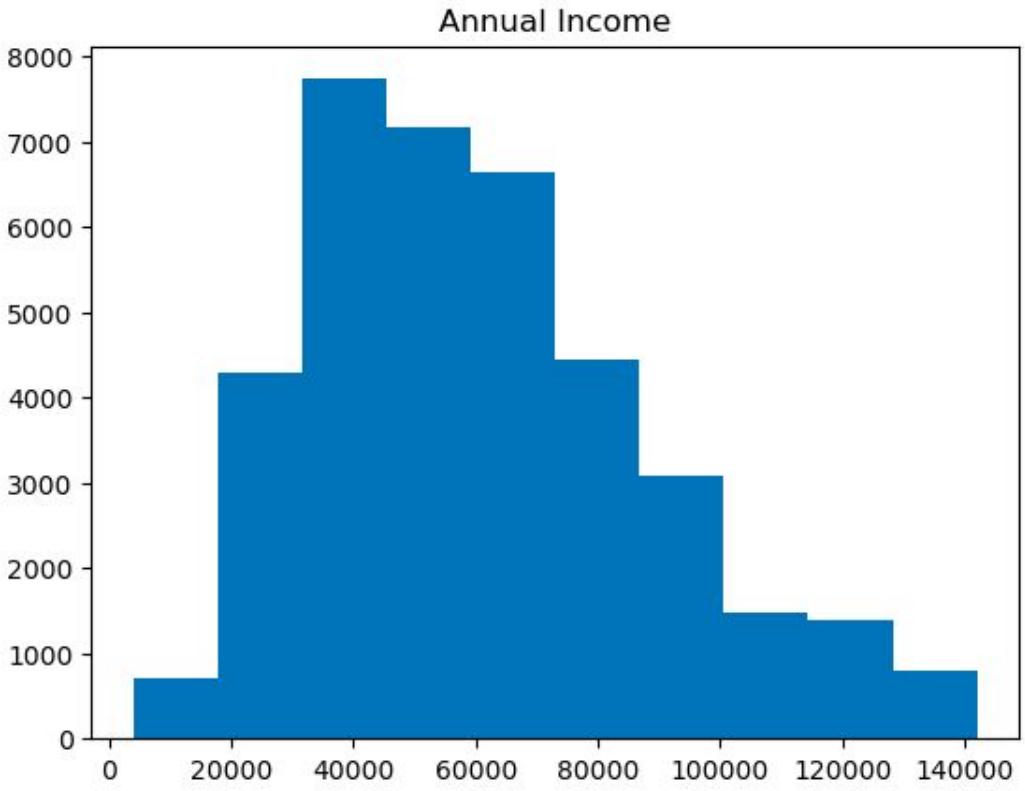
The distribution of Loan status show that the majority of loans is in the "Fully Paid" category, with a significant number in the "Charged Off" category, indicating default. This is critical for identifying patterns related to defaults.

Loan Amount Distribution



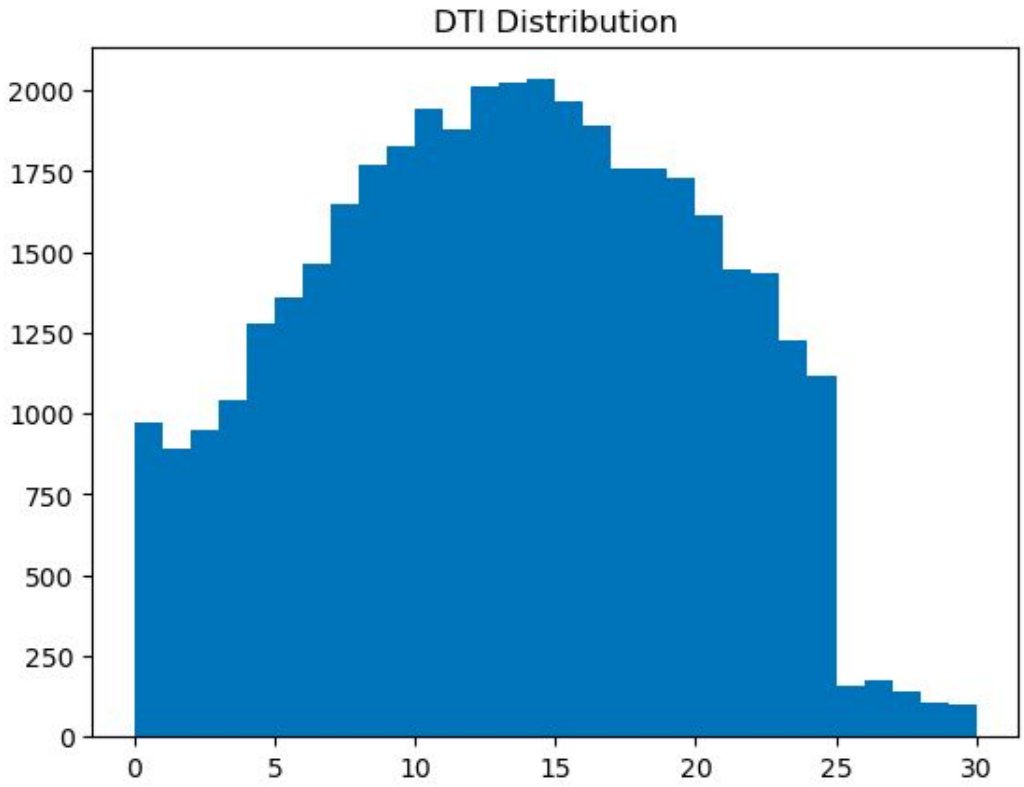
The distribution of loan amounts shows a right-skewed pattern, indicating that smaller loan amounts are more common than larger ones.

Annual Income Distribution



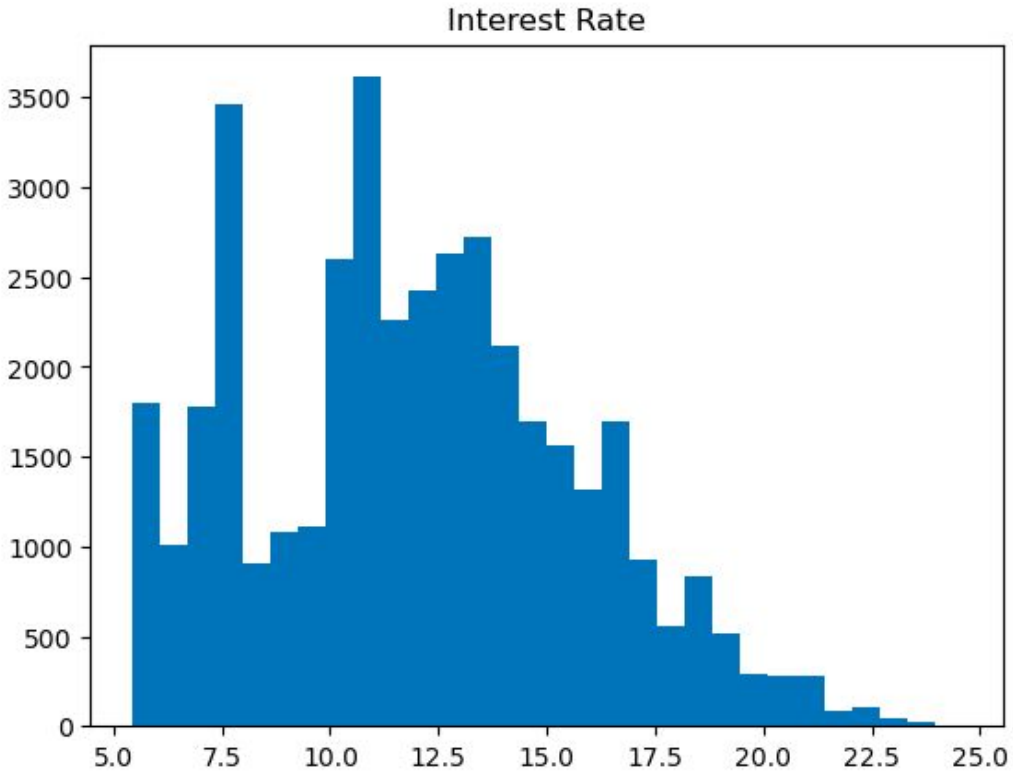
The distribution of annual income shows that most borrowers have reported incomes in the lower to middle range.

DTI (Debt-to-Income Ratio) Distribution



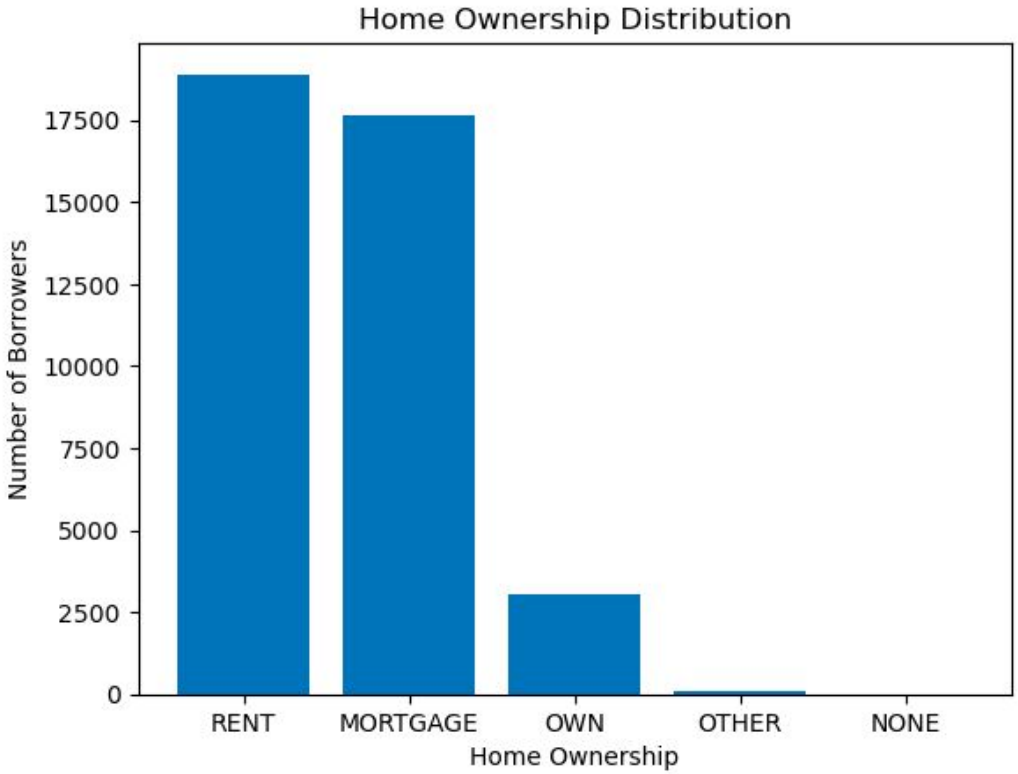
The DTI ratio has a relatively uniform distribution across a wide range of values, with a slight right skew.

Interest Rate Distribution:



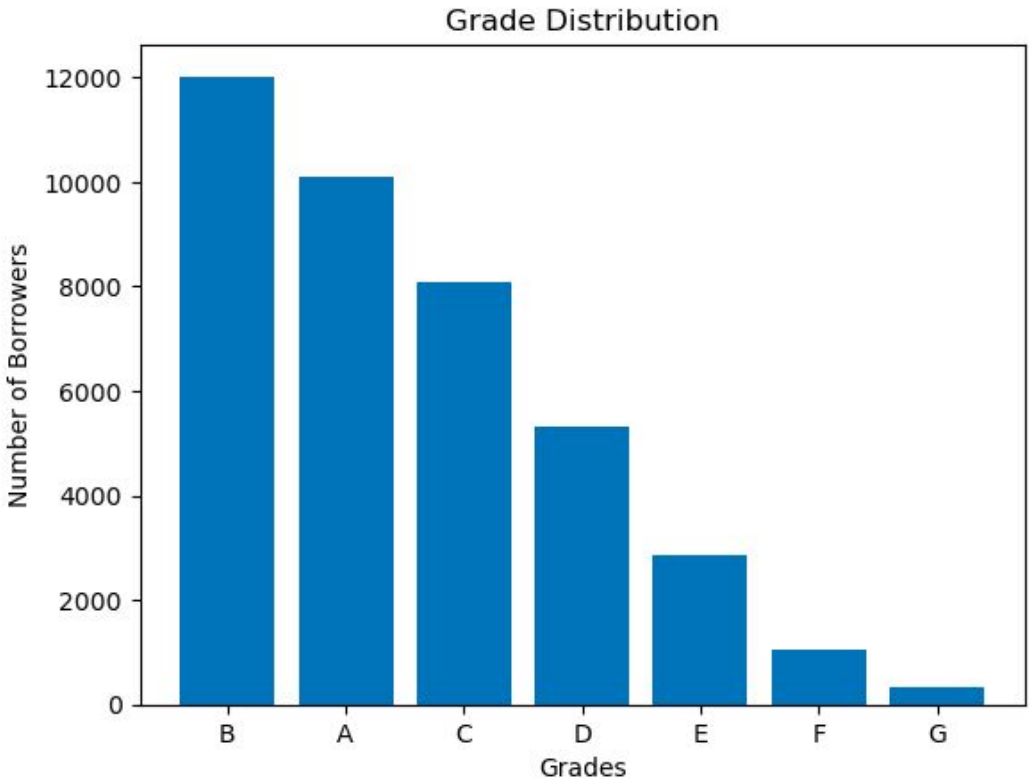
The interest rate distribution does not readily indicate any conventional pattern, reflecting different risk categories assigned to loans

Home Ownership Distribution



The distribution of the home ownership show that most of borrowers rent their home or have a mortgage against the home. Only a small number have indicate own home ownership

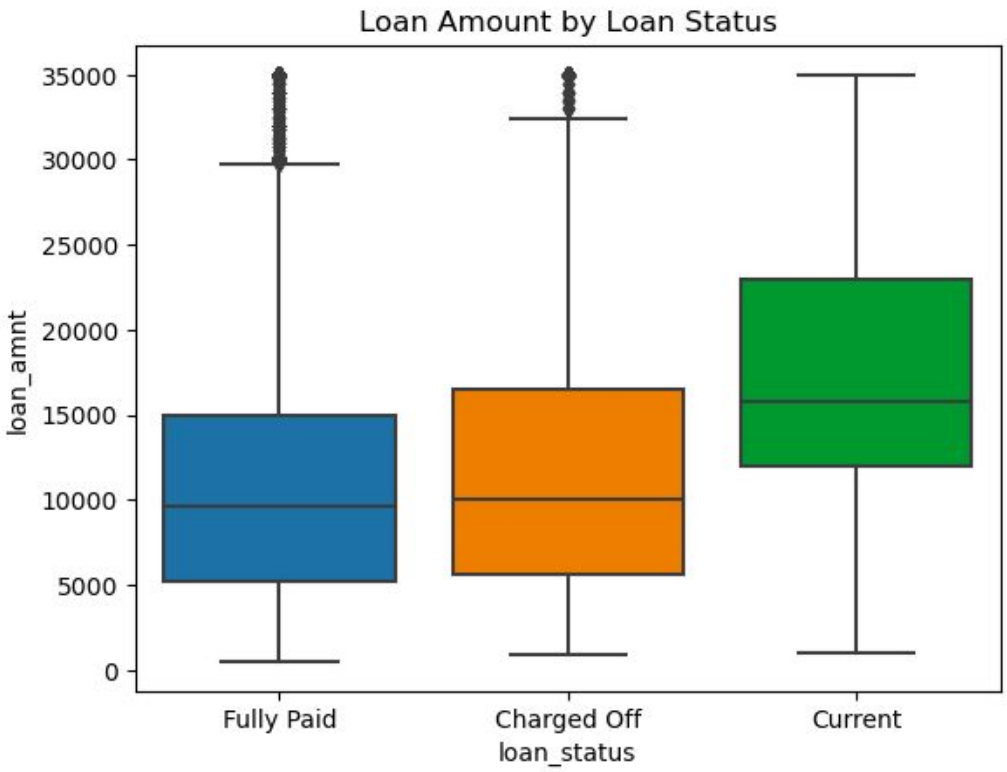
Grade Distribution



The distribution of grades show that most borrowers are assigned in the grade B, followed by A, C, D and so on

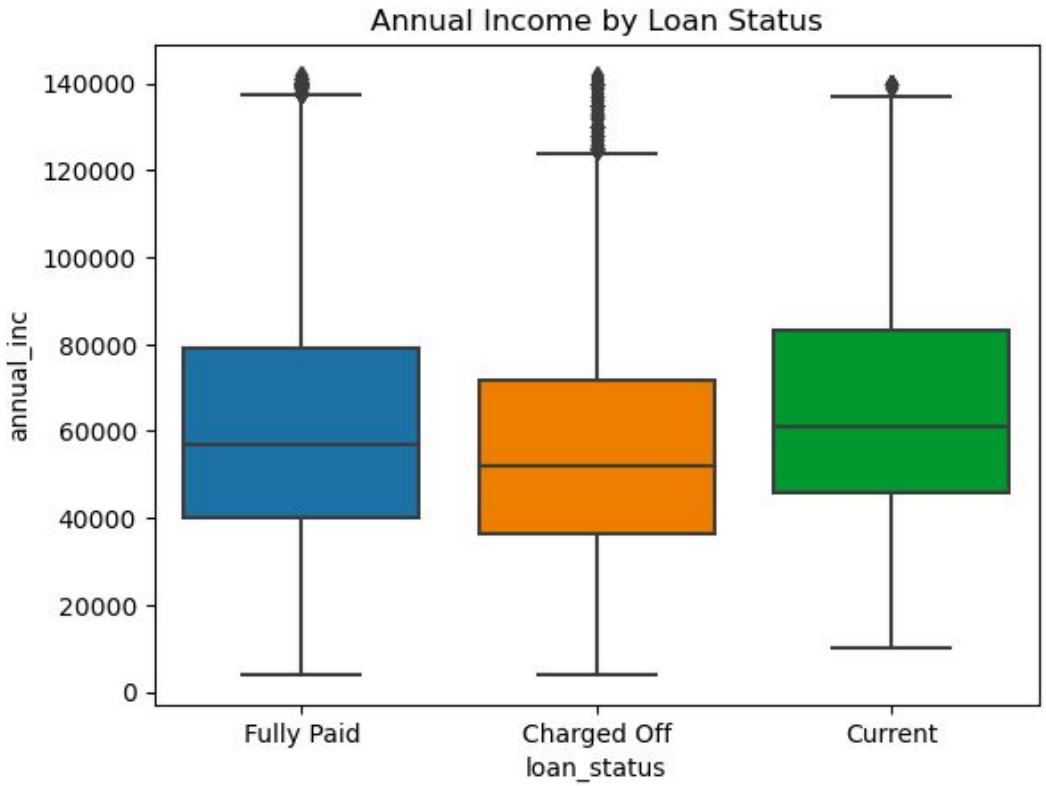
Segmented Univariate Analysis

Loan Amount vs. Loan Status



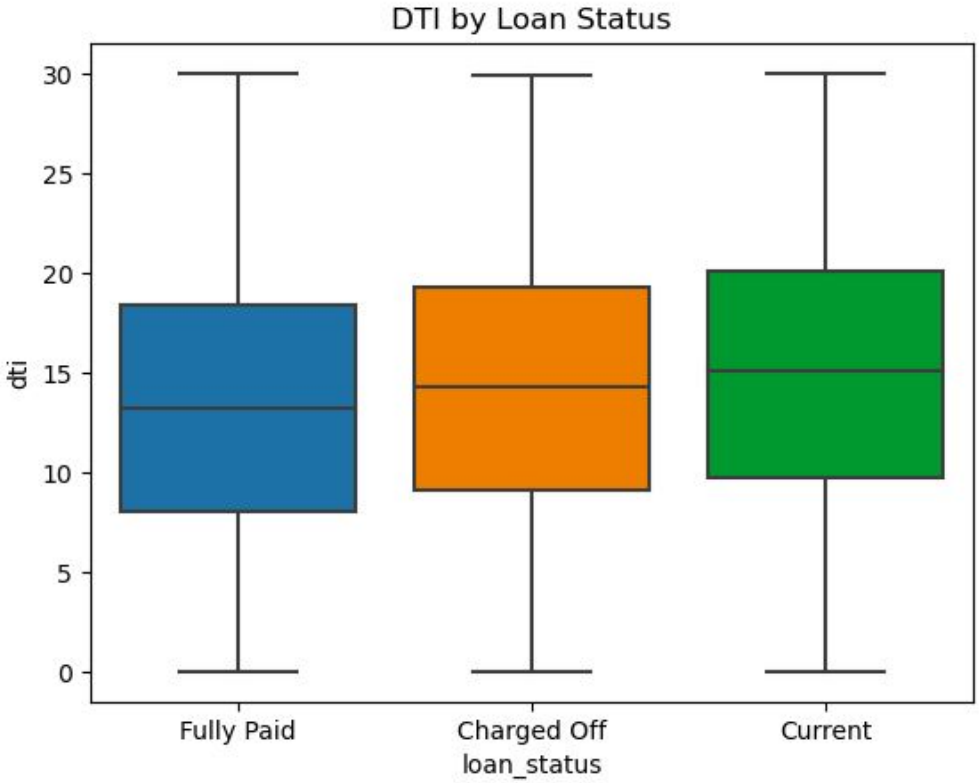
The median loan amount for charged-off loans is slightly higher than for fully paid loans, suggesting that higher loan amounts may have a slightly increased risk of default. However, the difference is not substantial, indicating other factors also play a significant role in the likelihood of default.

Annual Income vs. Loan Status



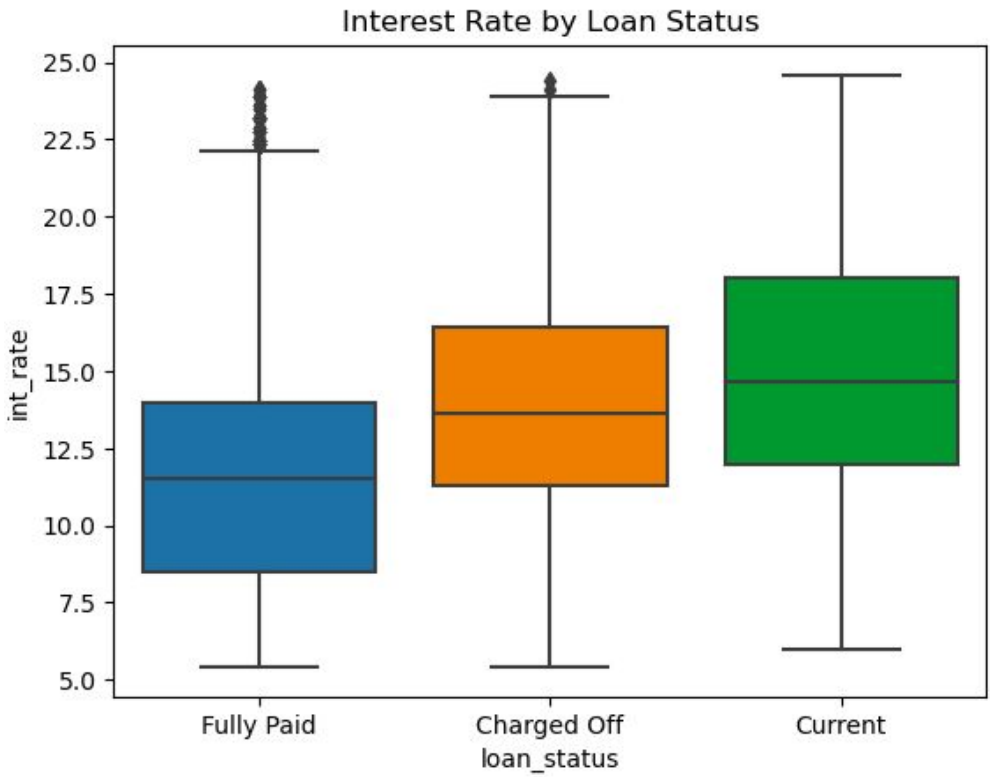
The median annual income appears to be lower for charged-off loans compared to fully paid loans, though the difference is not very pronounced. This suggests that borrowers with lower incomes might be at a higher risk of default, but income alone is not a definitive predictor.

DTI vs. Loan Status



The median DTI ratio is higher for charged-off loans than for fully paid loans. This indicates that a higher debt-to-income ratio is associated with an increased risk of default, which aligns with the expectation that borrowers with higher financial obligations relative to their income are more likely to face difficulties in repaying their loans.

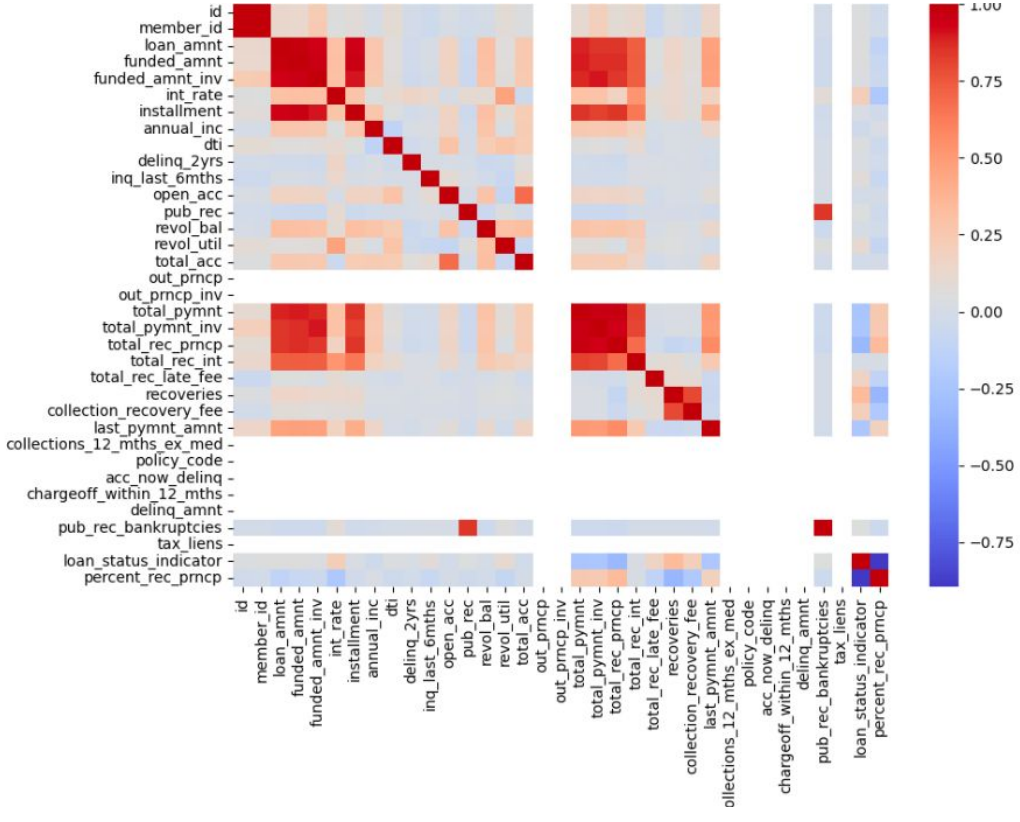
Interest Rate vs. Loan Status



There is a noticeable difference in the interest rates between fully paid and charged-off loans, with charged-off loans having higher median interest rates. This suggests that loans with higher interest rates, indicative of higher risk as assessed by the lender, are more likely to default.

Bivariate Analysis

Correlation analysis of loan status with other numerical features

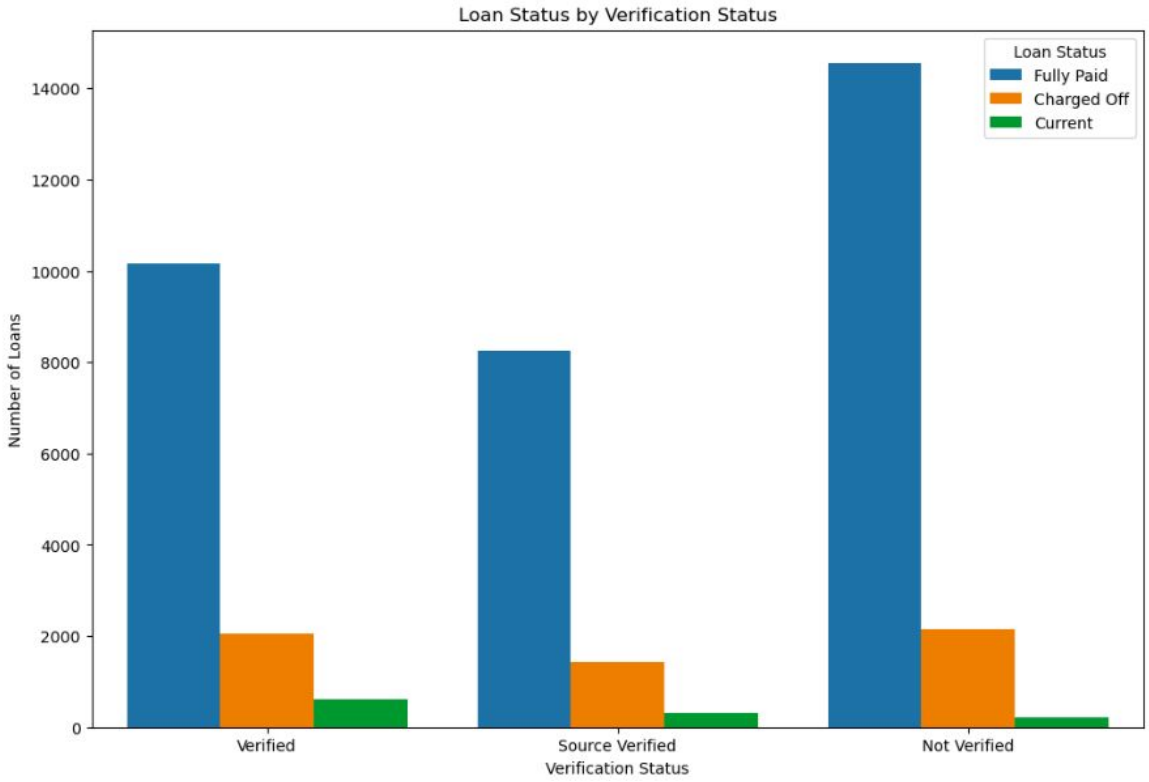


loan_status_indicator	1.000000
percent_rec_prncp	-0.893785
recoveries	0.339562
total_rec_prncp	-0.334944
total_pymnt	-0.235898
total_pymnt_inv	-0.232906
last_pymnt_amnt	-0.222524
int_rate	0.211390
collection_recovery_fee	0.204914
total_rec_late_fee	0.165115
revol_util	0.099875
inq_last_6mths	0.071878
loan_amnt	0.059463
funded_amnt	0.056497
pub_rec	0.051001
pub_rec_bankruptcies	0.047757
dti	0.045078
id	0.042390
annual_inc	-0.040867
member_id	0.039798
funded_amnt_inv	0.037781
installment	0.027153
total_acc	-0.022608
delinq_2yrs	0.020096
total_rec_int	0.013008
open_acc	-0.009140
revol_bal	0.005854

Observations of Correlation Analysis

1. Interest Rate (int_rate): Higher interest rates may correlate with a higher likelihood of default, as they indicate riskier loans.
2. Loan Amount (loan_amnt): Larger loan amounts could have a higher default risk. Most probably due to the increased financial burden on the borrower.
3. Annual Income (annual_inc): Borrowers with higher annual incomes might be less likely to default, reflecting a negative correlation with the default indicator.
4. Debt-to-Income Ratio (dti): A higher DTI ratio might show a positive correlation with default risk, indicating that borrowers with a higher debt load relative to their income are more likely to default.
5. installment (installment) : A positive correlation between the installment and loan_status is observed indicating that a higher monthly payment installments may have a higher likelihood of default. This is most probably due to increased burden of payment on the borrowers.
6. Percent of Principal Paid (percent_rec_prncp): Borrowers who have paid higher percentage of loan amount are less likely to default. This can be explained by the fact that borrowers with higher payments have higher motivation to complete the payment and close the loan.

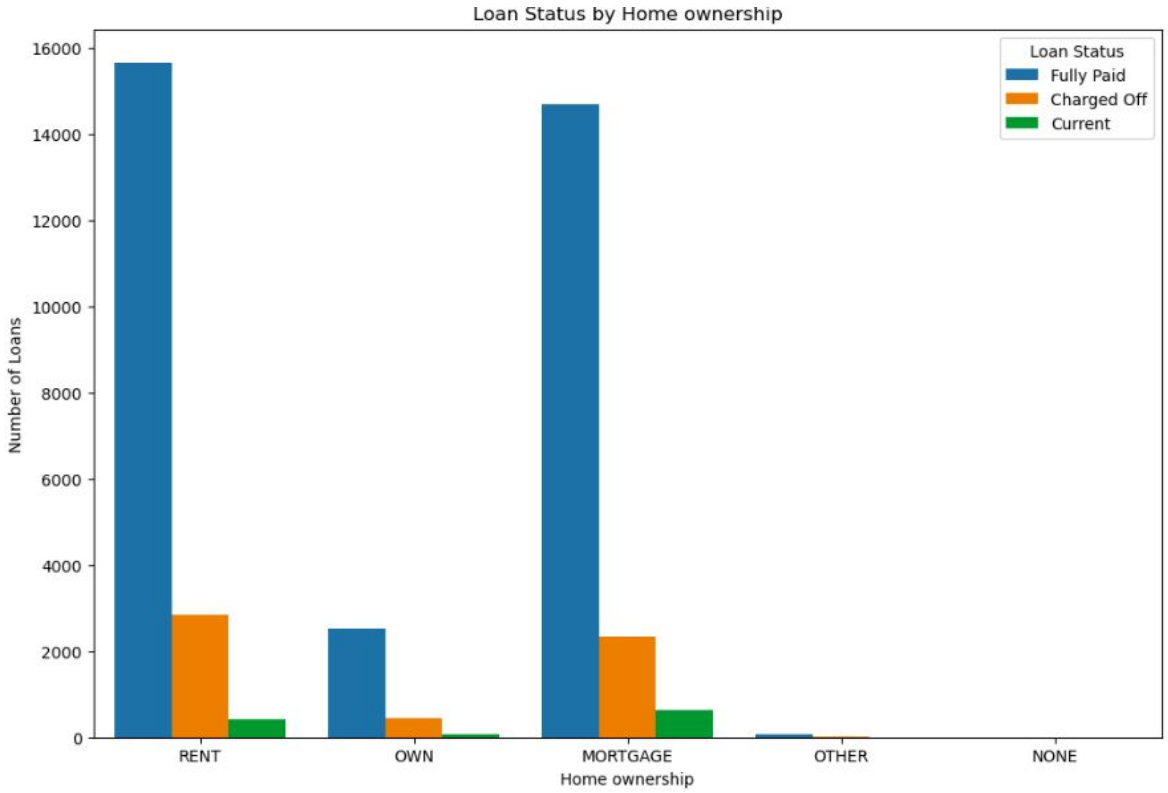
Relationship between Verification Status and Loan Status



	verification_status	Charge-Off Rate
0	Not Verified	0.126588
1	Source Verified	0.143587
2	Verified	0.160122

We observe higher delinquency in the verified borrowers

Relationship between Home Ownership and Loan Status



	home_ownership	Charge-Off Rate
0	MORTGAGE	0.131774
1	NONE	0.000000
2	OTHER	0.183673
3	OWN	0.144866
4	RENT	0.150220

We observe higher delinquency for people with Home Mortgages, as compared to borrowers who own or rent their homes

Recommendations

Recommendations

Based on these findings, to mitigate the risk of loan defaults, the company could consider the following actions:

1. DTI Ratios: Applicants with high DTI ratios could be subjected to more stringent review processes.
2. Interest Rate Adjustments: Higher interest rates correlate with higher default rates. Adjusting lending criteria or pricing for these loans could help balance risk and return.
3. Annual Income (annual_inc): Lending club should do additonal judgements of those borrowers with relatively lower annual incone due to higher risk of default.
4. Loan Amount Limits: Implementing stricter limits or higher scrutiny for larger loan amounts could also help manage default risk.

Ideally the company should target borrowers with higher income but for relatively lower income borrowers loan default risk can be mitigated for them by providing lower loan amounts, have lower installments, lower interests etc